

has made it possible to change the ways of living and how societies are organised and therefore have a great influence on the development of civilisations. Modern civilisation is totally dependent on different modes of transportation for not only traveling to different places but also for trading economic resources like raw materials, fuels, food, and manufactured goods from their place of origin to place of use.

Pollution problem

The use of a bicycle, which is currently the only human-friendly, cost-effective, and pollution-free mode of transportation, has reduced just to an exercise device or for leisure trips. Transportation is a major source of air pollution as cars, trucks, and buses powered by fossil fuels are major contributors to air pollution. According to pollution studies, it is said that the transportation sector contributes to more than half of nitrogen oxides into the air and is a major source of global warming emissions.

Further, studies have also linked pollutants from vehicle exhaust to have adverse impacts on nearly every organ in the body. With the so-called progress of civilisation or modernisation, not only various types of transportation vehicles like automobiles (motorcycles, scooters, and cars), buses, trains, trucks, helicopters, watercraft, spacecraft, and aircraft have flooded the market, but also their general and personal purchase and use have reached a stage where they have become the biggest sources of unwanted environmental pollution.

Going electric

Feeling concerned with degrading environmental conditions due to the pollution being generated from diesel and petrol-based transportation modes, several companies and governments have come forward with the idea of developing and using electric vehicles (EVs).

This shift was also possible due to reduced terminations, reconnections and arrearages, more stable electric energy prices, a more resilient and reliable electric energy system, lower transmission and distribution costs, and utility insurance savings.



TWIKE is the world's oldest only human electric hybrid car (Credit: <https://upload.wikimedia.org>)

There are a number of benefits of using EVs over conventional petrol/diesel vehicles like these are cheaper to run and maintain, better for the environment, use eco-friendly materials, and many more. The growing need for energy and degrading environmental considerations in the last few years have given direction to new paradigms of mobility and transportation with an increased interest in light electric vehicles (LEVs). In this context, electrically power-assisted cycles (EPACs) have received great attention. Several efforts have been made in order to improve their performance in terms of autonomy, weight, aesthetics, and feeling with the driver.

Technological developments taking place over the last few decades towards designing and developing efficient, long-lasting, and cost-effective batteries have also strengthened the move towards the emergence of EVs in the market. At present, only two-wheelers like scooters and motorcycles have been successful as electric vehicles, but it is expected that in the coming times, we will have fully electric buses, trucks, trains, and airplanes as well.

With the advanced batteries having more charge capacity, long life, and light weight, this will be

possible. Over the last two decades, lithium-ion battery technology has worked its way to the forefront of the automotive market. But EV battery technology continually pushes the boundaries of lithium-ion batteries' ability to provide ample power, longevity, and safety.

Fully battery-powered electric vehicles (BEV) rely exclusively on electric power to drive, while plug-in hybrid electric vehicles (PHEVs) and full hybrid electric vehicles (FHEV) work in tandem with ICEs to generate and provide power. All these vehicle types require different levels of battery complexity and size. Different EV vehicles demand different battery capacities.

While designing EV battery systems, engineers also consider charging speed, charge cycle ability, degradation, chemistry, and safety. Energy and power density thresholds have been realised in most EV applications, yet vehicle manufacturers are constantly tweaking module and cell sizes for optimum performance levels. Regardless of the lithium-ion battery cell and module sizes, the high-voltage battery systems that power EVs require meticulously designed battery management systems (BMS) to ensure maximum power and safety.